Math 261, Homework Set 4

Evaluate the following double integrals, showing all of your work (you do NOT need to simplify your answers).

1. $\int_0^1 \int_3^5 x^2 y \, dx \, dy$

2. $\int_1^4 \int_0^5 y \cos(xy) \, dx \, dy$

3. $\int_0^1 \int_0^{y^2} y e^x \, dx \, dy$

4. $\int_0^2 \int_y^{y^2} 15y \, dx \, dy$

5. $\int_0^1 \int_0^{\pi/4} e^y \sec^2(x) \, dx \, dy$

6. $\int_1^2 \int_x^{e^x} xy \, dy \, dx$

7. $\int_0^2 \int_0^{2x} xe^{x^3}\, dy\, dx$

8. $\int_0^1 \int_1^{1+x^2} \frac{x}{\sqrt{y}}\, dy\, dx$

9. $\int_0^{\pi} \int_0^1 x \sin(y)\, dx\, dy$

Evaluate the following triple integrals, showing all of your work (you do NOT need to simplify your answers).

10. $\int_0^1 \int_0^2 \int_{-1}^1 x^2yz^4 \, dz \, dy \, dx$

11. $\int_0^1 \int_0^x \int_0^2 3 \, dz \, dy \, dx$

12. $\int_0^1 \int_0^1 \int_y^z x^2 \, dx \, dy \, dz$

13. $\int_0^1 \int_{-1}^1 \int_0^{(1+z^3)^5} \frac{z^2}{y+9}\, dx\, dy\, dz$

14. $\int_1^2 \int_0^1 \int_0^{xy} e^{z/y}\, dz\, dy\, dx$

15. $\int_0^1 \int_0^x \int_0^1 \frac{z}{(1+x^2+z^2)^2}\, dz\, dy\, dx$

Evaluate the following integrals, showing all of your work, using the method of changing the order of integration (you do NOT need to simplify your answers)

16. $\int_0^1 \int_{3y}^3 e^{x^2}\, dx\, dy$

17. $\int_0^{\sqrt{\pi}} \int_x^{\sqrt{\pi}} \cos(y^2)\, dy\, dx$

18. $\int_0^{16} \int_{y^{1/4}}^2 \sin(x^5)\, dx\, dy$

19. $\int_0^1 \int_{\sqrt{x}}^1 \sqrt{2+y^3}\, dy\, dx$

20. $\int_0^3 \int_{y^2}^9 y \sin(x^2)\, dx\, dy$

21. $\int_0^1 \int_{e^x}^e \frac{y}{\ln(y)}\, dy\, dx$

22. Use a change in the order of integration to evaluate the integral sum $\int_0^e \int_1^2 \sec^2(e^x)\,dx\,dy + \int_e^{e^2} \int_{\ln(y)}^2 \sec^2(e^x)\,dx\,dy$.

23. Evaluate the triple integral $\int_{-1}^1 \int_0^1 \int_y^1 e^{x^2} z^2\,dx\,dy\,dz$.

24. Evaluate the triple integral $\int_0^1 \int_0^{\sqrt{z}} \int_{x^2}^z \sqrt{z} e^{y^{3/2}}\,dy\,dx\,dz$.

25. Evaluate the triple integral $\int_0^1 \int_0^z \int_y^z z\cos(x^2)\,dx\,dy\,dz$.

26. Use a change in the order of integration to evaluate the integral sum $\int_0^1 \int_0^{z^2} \int_z^2 z^2 e^{x^3}\,dx\,dy\,dz + \int_0^1 \int_{z^2}^4 \int_{\sqrt{y}}^2 z^2 e^{x^3}\,dx\,dy\,dz$.

27. Which of the following integrals are set up in a way that makes mathematical sense? (There may be multiple correct answers).

    (a) $\int_0^1 \int_1^3 \int_x^y f(x,y,z)\,dx\,dy\,dz$
    (b) $\int_0^1 \int_z^1 \int_y^{2y} f(x,y,z)\,dx\,dy\,dz$
    (c) $\int_0^z \int_0^1 \int_0^y f(x,y,z)\,dx\,dy\,dz$
    (d) $\int_{-1}^1 \int_z^y \int_0^1 f(x,y,z)\,dx\,dy\,dz$
    (e) $\int_0^1 \int_0^y \int_x^y f(x,y,z)\,dz\,dx\,dy$
    (f) $\int_0^1 \int_0^z \int_x^y f(x,y,z)\,dz\,dx\,dy$
    (g) $\int_0^1 \int_0^1 \int_0^{xy} f(x,y,z)\,dz\,dy\,dx$
    (h) $\int_0^1 \int_0^1 \int_0^{xyz} f(x,y,z)\,dz\,dy\,dx$
    (i) $\int_0^z \int_0^x \int_0^y f(x,y,z)\,dz\,dy\,dx$
    (j) $\int_0^1 \int_{z^2}^{e^z} \int_{yz}^{y^2z^2} f(x,y,z)\,dx\,dy\,dz$